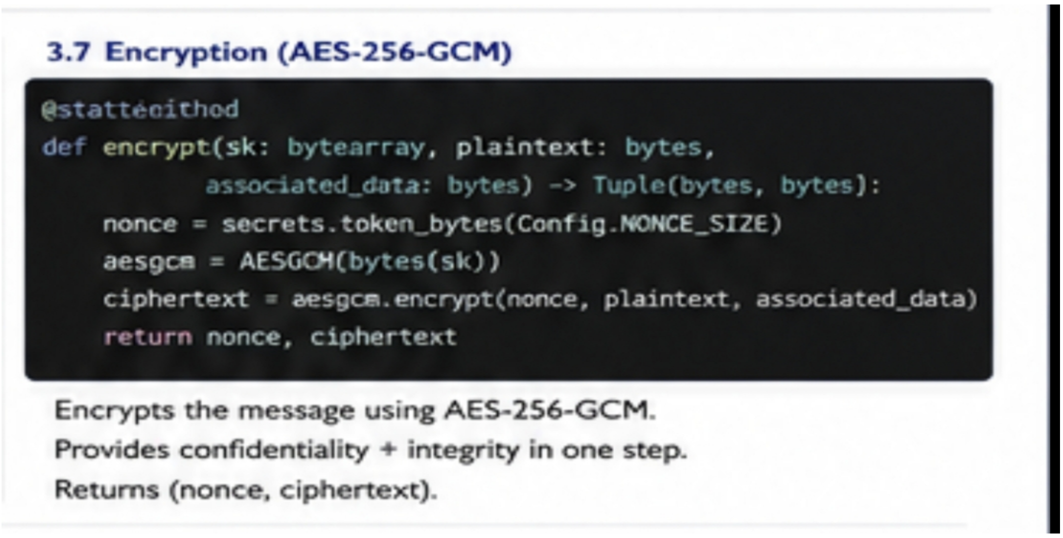
PRODUCTION READY CODE OF X3DH

1. Overview

X3DH (Extended Triple Diffie-Hellman) establishes a secure channel between Alice and Bob. This implementation uses X25519, Ed25519, HKDF-SHA256 and AES-256-GCM.

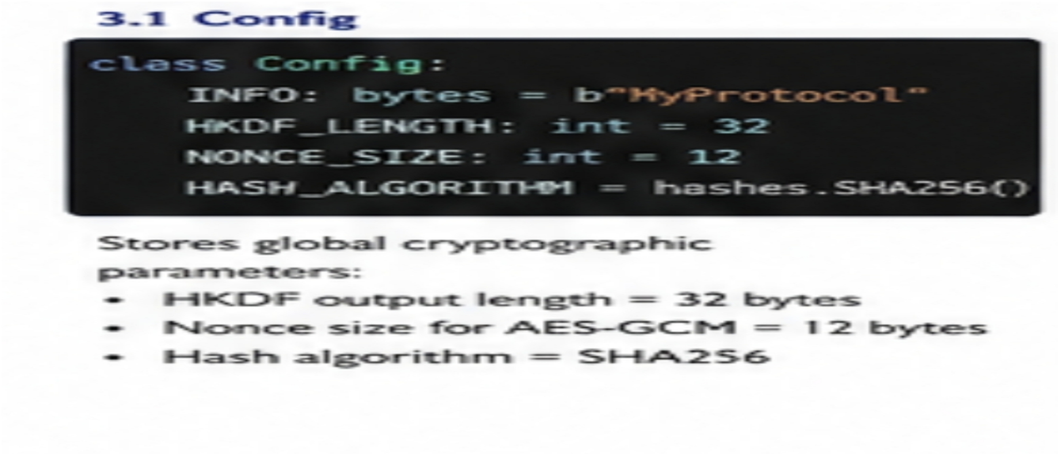


3.7 Encryption (AES-256-GCM)

```python
@staticmethod
def encrypt(sk: bytearray, plaintext: bytes,
            associated_data: bytes) -> Tuple(bytes, bytes):
    nonce = secrets.token_bytes(Config.NONCE_SIZE)
    aesgcm = AESGCM(bytes(sk))
    ciphertext = aesgcm.encrypt(nonce, plaintext, associated_data)
    return nonce, ciphertext
```

Encrypts the message using AES-256-GCM.
Provides confidentiality + integrity in one step.
Returns (nonce, ciphertext).

Config stores global parameters such as nonce size, HKDF length and SHA256 algorithm.



3.1 Config

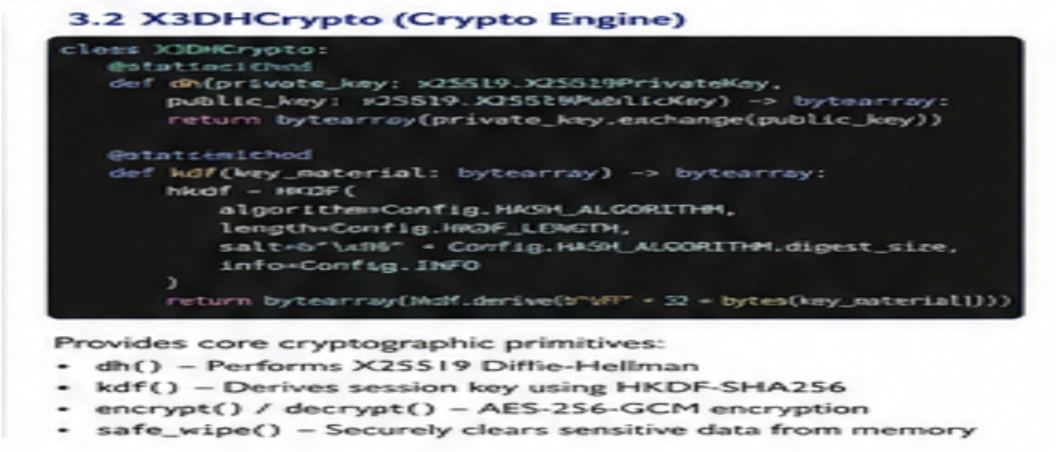
```python
class Config:
    INFO: bytes = b"MyProtocol"
    HKDF_LENGTH: int = 32
    NONCE_SIZE: int = 12
    HASH_ALGORITHM = hashes.SHA256()
```

Stores global cryptographic parameters:

- HKDF output length = 32 bytes
- Nonce size for AES-GCM = 12 bytes
- Hash algorithm = SHA256

X3DHCrypto performs Diffie-Hellman, HKDF derivation, encryption and secure memory wiping.

2. Secure Components



3.2 X3DHCrypto (Crypto Engine)

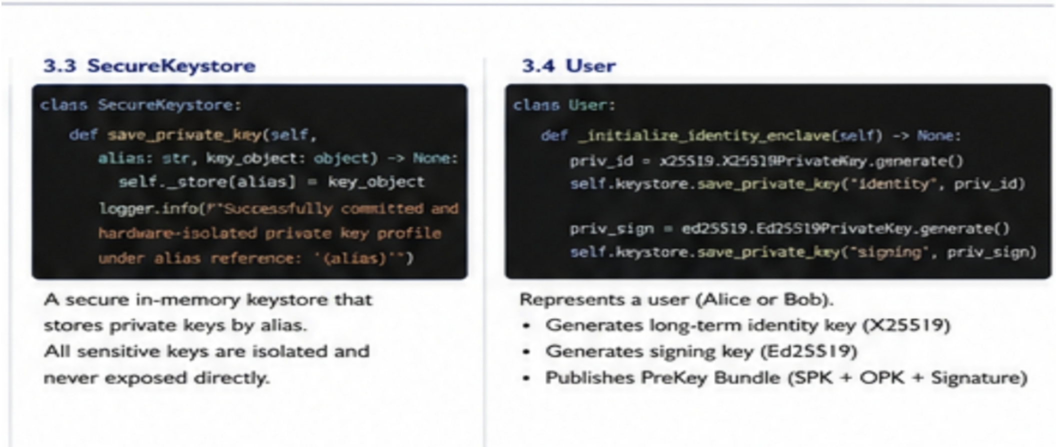
```python
class X3DHCrypto:
    @staticmethod
    def dh(private_key: x25519.X25519PrivateKey,
           public_key: x25519.X25519PublicKey) -> bytearray:
           return bytearray(private_key.exchange(public_key))

    @staticmethod
    def kdf(key_material: bytearray) -> bytearray:
           hkdf = HKDF(
               algorithm=Config.HASH_ALGORITHM,
               length=Config.HKDF_LENGTH,
               salt=b"\x00" * Config.HASH_ALGORITHM.digest_size,
               info=Config.INFO
           )
           return bytearray(hkdf.derive(b"\xFF" * 32 + bytes(key_material)))
```

Provides core cryptographic primitives:

- dh() — Performs X25519 Diffie-Hellman
- kdf() — Derives session key using HKDF-SHA256
- encrypt() / decrypt() — AES-256-GCM encryption
- safe_wipe() — Securely clears sensitive data from memory

SecureKeystore safely stores private keys. User creates identity keys and publishes prekey bundles.



3.3 SecureKeystore

```python
class SecureKeystore:
    def save_private_key(self,
        alias: str, key_object: object) -> None:
          self._store[alias] = key_object
        logger.info(f"Successfully committed and
        hardware-isolated private key profile
        under alias reference: '{alias}'")
```

A secure in-memory keystore that stores private keys by alias. All sensitive keys are isolated and never exposed directly.

3.4 User

```python
class User:
    def _initialize_identity_enclave(self) -> None:
        priv_id = x25519.X25519PrivateKey.generate()
        self.keystore.save_private_key("identity", priv_id)

        priv_sign = ed25519.Ed25519PrivateKey.generate()
        self.keystore.save_private_key("signing", priv_sign)
```

Represents a user (Alice or Bob).

- Generates long-term identity key (X25519)
- Generates signing key (Ed25519)
- Publishes PreKey Bundle (SPK + OPK + Signature)

PreKeyServer stores bundles and prevents replay attacks using nonce tracking.

3. X3DH Session Analysis

3.5 PreKeyServer

```
class PreKeyServer:
    def verify_nonce_freshness(self, nonce: bytes) -> None:
        if nonce in self._used_nonces:
            raise SecurityError(
                "Replay attack verified: Provided
                packet inttialization vector was
                already marked dead."
            )
        self._used_nonces.add(nonce)
```

Acts as a directory server storing PreKey Bundles.
It also tracks used nonces to prevent replay attacks.

X3DHSession performs DH1, DH2, DH3 and DH4 calculations and derives a common session key.

3.6 X3DHSession

```
class X3DHSession:
    def initiator(self, sender: User, recipient_id: str,
              server: PreKeyServer, peer_signing_pub: ed25519.Ed25519PublicKey):
        # Performs 4 DH calculations
        dh1 = X3DHCrypto.dh(sender_id_priv, bundle.signed_prekey_public)
        dh2 = X3DHCrypto.dh((e_priv, bundle.identity_public)
        dh3 = X3DHCrypto.dh((e_priv, bundle.signed_prekey_public)
        dh4 = X3DHCrypto.dh((e_priv, bundle.one_time_prekey_public)
        # Derive session key
        self.session_key = X3DHCrypto.kdf(dh1 + dh2 + dh3 + dh4)
```

Contains initiator() and responder() methods.
Both sides perform the same 4 DH operations and derive the same session key.

AES-256-GCM encrypts and authenticates the message simultaneously.

4. Workflow Summary

• 1. Bob publishes Signed Prekey and One-Time Prekey.

• 2. Alice fetches Bob's bundle.

• 3. Alice verifies Bob's signature.

• 4. Alice computes 4 DH calculations.

• 5. HKDF derives a 32-byte session key.

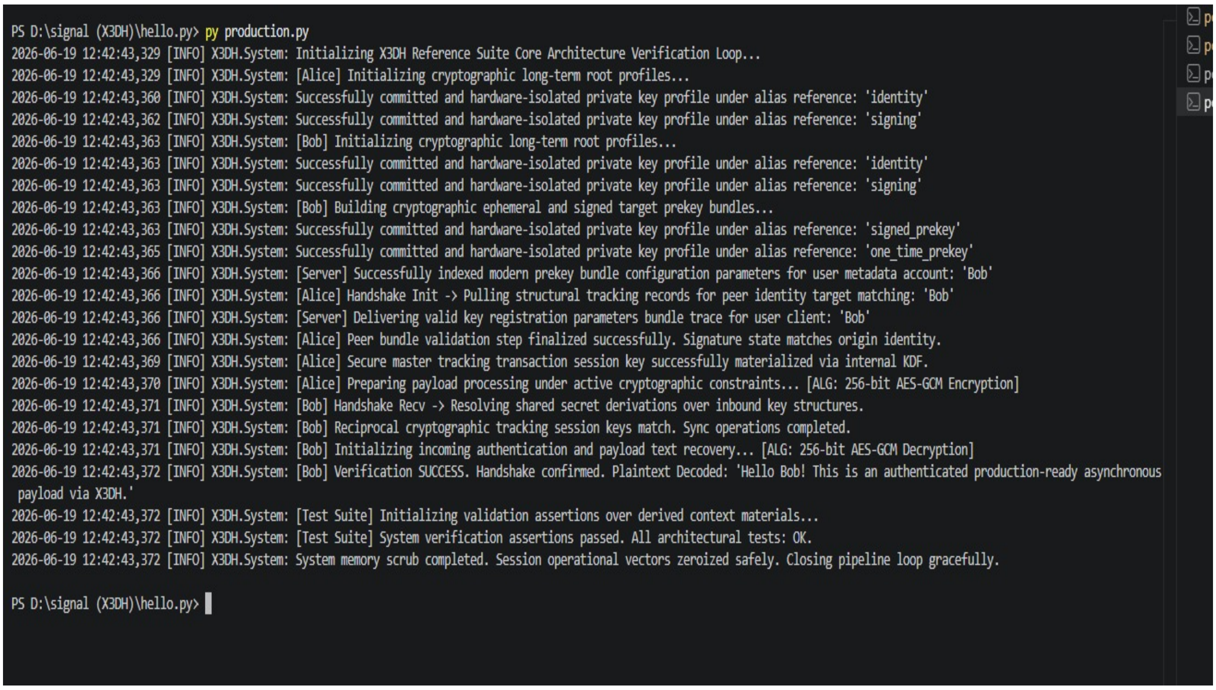
• 6. Alice encrypts the message.

• 7. Bob derives the same key and decrypts the message.

• 8. Session keys are securely wiped from memory.

Security Features: Mutual Authentication, Forward Secrecy, Replay Attack Protection, Secure Memory Wiping and AES-GCM Integrity Protection.

5. Terminal Output



```
PS D:\signal (X3DH)\hello.py> py production.py
2026-06-19 12:42:43,329 [INFO] X3DH.System: Initializing X3DH Reference Suite Core Architecture Verification Loop...
2026-06-19 12:42:43,329 [INFO] X3DH.System: [Alice] Initializing cryptographic long-term root profiles...
2026-06-19 12:42:43,360 [INFO] X3DH.System: Successfully committed and hardware-isolated private key profile under alias reference: 'identity'
2026-06-19 12:42:43,362 [INFO] X3DH.System: Successfully committed and hardware-isolated private key profile under alias reference: 'signing'
2026-06-19 12:42:43,363 [INFO] X3DH.System: [Bob] Initializing cryptographic long-term root profiles...
2026-06-19 12:42:43,363 [INFO] X3DH.System: Successfully committed and hardware-isolated private key profile under alias reference: 'identity'
2026-06-19 12:42:43,363 [INFO] X3DH.System: Successfully committed and hardware-isolated private key profile under alias reference: 'signing'
2026-06-19 12:42:43,363 [INFO] X3DH.System: [Bob] Building cryptographic ephemeral and signed target prekey bundles...
2026-06-19 12:42:43,363 [INFO] X3DH.System: Successfully committed and hardware-isolated private key profile under alias reference: 'signed_prekey'
2026-06-19 12:42:43,365 [INFO] X3DH.System: Successfully committed and hardware-isolated private key profile under alias reference: 'one_time_prekey'
2026-06-19 12:42:43,366 [INFO] X3DH.System: [Server] Successfully indexed modern prekey bundle configuration parameters for user metadata account: 'Bob'
2026-06-19 12:42:43,366 [INFO] X3DH.System: [Alice] Handshake Init -> Pulling structural tracking records for peer identity target matching: 'Bob'
2026-06-19 12:42:43,366 [INFO] X3DH.System: [Server] Delivering valid key registration parameters bundle trace for user client: 'Bob'
2026-06-19 12:42:43,366 [INFO] X3DH.System: [Alice] Peer bundle validation step finalized successfully. Signature state matches origin identity.
2026-06-19 12:42:43,369 [INFO] X3DH.System: [Alice] Secure master tracking transaction session key successfully materialized via internal KDF.
2026-06-19 12:42:43,370 [INFO] X3DH.System: [Alice] Preparing payload processing under active cryptographic constraints... [ALG: 256-bit AES-GCM Encryption]
2026-06-19 12:42:43,371 [INFO] X3DH.System: [Bob] Handshake Recv -> Resolving shared secret derivations over inbound key structures.
2026-06-19 12:42:43,371 [INFO] X3DH.System: [Bob] Reciprocal cryptographic tracking session keys match. Sync operations completed.
2026-06-19 12:42:43,371 [INFO] X3DH.System: [Bob] Initializing incoming authentication and payload text recovery... [ALG: 256-bit AES-GCM Decryption]
2026-06-19 12:42:43,372 [INFO] X3DH.System: [Bob] Verification SUCCESS. Handshake confirmed. Plaintext Decoded: 'Hello Bob! This is an authenticated production-ready asynchronous
 payload via X3DH.'
2026-06-19 12:42:43,372 [INFO] X3DH.System: [Test Suite] Initializing validation assertions over derived context materials...
2026-06-19 12:42:43,372 [INFO] X3DH.System: [Test Suite] System verification assertions passed. All architectural tests: OK.
2026-06-19 12:42:43,372 [INFO] X3DH.System: System memory scrub completed. Session operational vectors zeroized safely. Closing pipeline loop gracefully.

PS D:\signal (X3DH)\hello.py>
```

The terminal output confirms successful initialization, key generation, encryption, decryption, unit testing and secure cleanup of memory.

Conclusion

The Production Ready X3DH implementation successfully establishes a secure communication channel between Alice and Bob.